



Luca Lazzaroni

Assistant Professor in Electronics

DITEN – UNIGE

BASED IN Genoa, Italy

EMAIL luca.lazzaroni@unige.it

SECTOR RTD-A · IINF-01/A

BIBLIOMETRICS			SCOPUS
65	329	11	67
SCOPUS DOCUMENTS	CITATIONS	H-INDEX	CO-AUTHORS

SYNCED 21 SEPT 2026

01 Profile

Assistant Professor (RTD-A) at the Department of Naval, Electrical, Electronic and Telecommunications Engineering (DITEN), University of Genoa, sector IINF-01/A – Electronics, since March 2025. PhD in Science and Technology for Electronic and Telecommunication Engineering since 2024.

Research on embedded machine learning and edge AI: TinyML toolchains, quantized and binary neural networks on microcontrollers and FPGAs, perception and decision-making for automated driving, deep reinforcement learning, on-device LLMs. The work is also carried out within European and national research projects with industrial partners.

- Edge AI
- TinyML
- Embedded deep learning
- Automated driving
- Deep reinforcement learning
- On-device LLMs

02 Appointments & education

POSITIONS

Mar 2025 – present	Assistant Professor (RTD-A) DITEN, University of Genoa Scientific sector IINF-01/A — Electronics.
Nov 2023 – Feb 2025	Research Fellow DITEN, University of Genoa Grant: “Machine Learning models for the processing of data from autonomous vehicles”.
Nov 2020 – Oct 2023	PhD Candidate DITEN, University of Genoa — XXXVI cycle Scholarship: “Deep learning on the edge”.

EDUCATION

2024	PhD in Science and Technology for Electronic and Telecommunication Engineering University of Genoa — awarded 14 March 2024 Thesis: “The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning”
2020	MSc in Electronic Engineering (LM-29) University of Genoa — 24 July 2020 Thesis: “Design and implementation of a multiplatform execution engine for IoT applications” · Grade: 110/110
2017	BSc in Electronic Engineering and Information Technologies (L-8) University of Genoa — 27 October 2017 Thesis: “Model-based design of Lego robot control in Simulink environment”
2014	Scientific High School Diploma Liceo Scientifico Statale “Arturo Issel”, Finale Ligure (SV) Grade: 100/100

Chartered Information Engineer — State Examination passed in the second session of 2020.

03 Research projects

Synergies	Generation of scenarios for development, training, virtual testing and validation of CCAM systems European — HORIZON-CL5-2023-D6-01-02 · Member of the University of Genoa research unit Generates test scenarios for training, testing and validating CCAM systems, focusing on urban and rural traffic — where relevant-scenario knowledge is far thinner than for highway driving. Runs with most major European carmakers and universities active in the field.
Hi-Drive	Addressing challenges toward the deployment of higher automation European — H2020, GA 101006664 · Member of the University of Genoa research unit Develops and tests advanced automated driving at scale, moving from conditional automation (SAE L3) to high automation across different scenarios and road contexts.
GEMINI	Green Machine Learning for the IoT National — PRIN 2022 · Member of the University of Genoa research unit (DII–UNITN / DITEN–UNIGE) A framework for fully sustainable TinyML applications deployed along an edge-to-cloud continuum, targeting battery-free embedded devices — including those with no network connectivity at all.
ADINA	Integrated Digital Assistant Automotive Regional — POR FESR Piedmont 2014-2020 · Member of the University of Genoa research unit, with Lifetouch An in-vehicle virtual assistant on dedicated hardware, connected to the car's systems: AI algorithms interpret the driver's requests and act on vehicle and infotainment functions.

04 Teaching & supervision

COURSES

A.Y. 2024/25, 2025/26	Distributed Electronic Systems and Technologies for Environmental Monitoring Course lecturer · cod. 94631
A.Y. 2024/25, 2025/26	Tiny Machine Learning Course lecturer · cod. 114752
since A.Y. 2022/23	Machine Learning for Automated Driving Subject expert · cod. 108716
A.Y. 2023/24	Computer Graphics Arch. — Games and Simulation Adjunct professor · cod. 172317
A.Y. 2022/23	Autonomous Agents in Games Teaching support · cod. 198216
A.Y. 2020/21	Fundamentals of Programming of Electronic Systems Teaching support · cod. 166061
A.Y. 2020/21	Makers Approach to Electronic Design Teaching support · cod. 186985

All courses at the University of Genoa.

SUPERVISION

- 1** PhD candidate — Ali Zare, Science and Technology for Electronic and Telecommunication Engineering (XLI cycle)
- 12** BSc theses — Electronic Engineering and Information Technologies (L-8)
- 10** MSc theses — Electronic Engineering (LM-29)
- 7** MSc theses — Technological Engineering for Strategy and Security (LM/DS)
- 2** MSc theses — Computer Science (LM-18)

05 Editorial & service

EDITORIAL ROLES

2025 **Editorial Board Member (Early Career)**
Sensors — MDPI
Since 18 December 2025.

2025	Volume Editor APPLEPIES 2025 — Springer LNEE Proceedings of the International Conference on Applications in Electronics Pervading Industry, Environment and Society.
2024	Guest Editor Energies — MDPI Special Issue “Tiny Machine Learning for Energy Applications”, April 2024 — August 2025.

CONFERENCE ORGANISATION

Program Chair APPLEPIES 2025 Turin, Italy, 11–12 September 2025.
Program Committee APPLEPIES Since 2023.
Program Committee GALA — Games and Learning Alliance Since 2023.
Session Chair APPLEPIES 2023 1st Short Oral session — Genoa, Italy, 28–29 September 2023.

ACADEMIC APPOINTMENTS

Member of the Teaching and Admissions Committee, MSc in Engineering for Natural Risk Management.

PEER REVIEW

JOURNALS	CONFERENCES
IEEE Access since 2023	IEEE ICRA — International Conference on Robotics and Automation since 2026
Journal of Systems Architecture since 2023	IEEE/RSJ IROS — Intelligent Robots and Systems since 2025
Engineering Applications of Artificial Intelligence since 2024	GALA — Games and Learning Alliance since 2023, 2024
Sensors since 2024	APPLEPIES since 2021–2024
IET Intelligent Transport Systems since 2025	
IEEE Transactions on Intelligent Vehicles since 2026	

IEEE Transactions on Circuits and Systems II: Express Briefs	since 2026
Robotics and Autonomous Systems	since 2026
Scientific Reports	since 2026
Journal of Big Data	since 2026
Discover Internet of Things	since 2026

06 Invited & conference talks

13 TALKS

11–12 Sep 2025	APPLEPIES 2025 Turin, Italy
11–14 Jan 2025	43rd IEEE ICCE — International Conference on Consumer Electronics Las Vegas, NV, USA
19–20 Sep 2024	APPLEPIES 2024 Turin, Italy
24–26 Jun 2024	55th Annual Meeting of the Italian Society of Electronics (SIE) Genoa, Italy
28–29 Sep 2023	APPLEPIES 2023 Genoa, Italy
7–9 Sep 2023	54th Annual Meeting of the Italian Society of Electronics (SIE) Noto (RG), Italy
26–27 Sep 2022	APPLEPIES 2022 Genoa, Italy
7–9 Sep 2022	53rd Annual Meeting of the Italian Society of Electronics (SIE) Pizzo (VV), Italy
21–22 Sep 2021	APPLEPIES 2021 Pisa, Italy
1–2 Dec 2021	GALA 2021 — Games and Learning Alliance La Spezia, Italy (online)
7–9 Jul 2021	52nd Annual Meeting of the Italian Society of Electronics (SIE) Trieste, Italy (online)
9–10 Dec 2020	GALA 2020 — Games and Learning Alliance Laval, France (online)

19–20 Nov 2020 **APPLEPIES 2020**
Online

07 Awards & memberships

2025 **Best Presentation of the Session — “Edge Computing”**
43rd IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 11–14 January 2025
A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti — “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices”

PROFESSIONAL MEMBERSHIPS

IEEE
IEEE Member.

IEEE Consumer Technology Society
Member of the “Machine Learning, Deep Learning and AI in CE (MDA)” Technical Committee.

SIE — Italian Society of Electronics
Member within the University of Genoa Research Unit since 2021, research area “Electronic systems and applications”.

08 Contact & profiles

EMAIL luca.lazzaroni@unige.it	UNIGE rubrica.unige.it	ORCID 0000-0001-8092-5473
SCOPUS 57220892898	GOOGLE SCHOLAR w_6a6sYAAAAJ	GITHUB @lucalazzaroni
LINKEDIN luca-lazzaroni		

09 Publications

71 ITEMS · 65 IN SCOPUS

Every figure here comes from Scopus (author ID 57220892898) and is refreshed automatically. The 6 national-conference and workshop items that Scopus does not index are listed below, but contribute to none of them.

2026 **A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction**
L. Forneris, R. Berta, M. Fresta, **L. Lazzaroni**, H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, F. Bellotti
IEEE Signal Processing Letters vol. 33, pp. 136–140 Journal
<https://doi.org/10.1109/lsp.2025.3638676>

FPGA Deployment of Extremely Quantized Models for Time-to-Lane-Change Prediction

A. Saad, R. Berta, M. Fresta, **L. Lazzaroni**, A. Pighetti, O. Srour, F. Tango, F. Bellotti

IEEE Transactions on Circuits and Systems II: Express Briefs vol. 73, no. 8, pp. 888–892

Journal

<https://doi.org/10.1109/tcsii.2026.3705103>

Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms

H. Rojhan, F. Bellotti, M. Fresta, **L. Lazzaroni**, R. Berta

IEEE Access vol. 14, pp. 44886–44903

Journal

<https://doi.org/10.1109/access.2026.3676567>

StabilEdge, Formal Verification of Edge AI Applications

L. Lazzaroni, R. Berta, V. Soltanmuradov, D. Martin Gomez, F. Bellotti

IEEE Access vol. 14, pp. 37416–37434

Journal

<https://doi.org/10.1109/access.2026.3667991>

Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers

A. Saad, F. Bellotti, H. Ballout, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 321–327

Conference

https://doi.org/10.1007/978-3-032-17174-0_46

Preface

M. Ruó Roch, R. Berta, A. Gómez, **L. Lazzaroni**, M. Martina, P.M. Ros, N. Pazos Escudero

Lecture Notes in Electrical Engineering pp. v–vi

Editorial

Optimizing Binary AutoEncoders on STM32 Microcontrollers

H. Ballout, R. Berta, **L. Lazzaroni**, F. Sakr, A. Pighetti, M. Fresta, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 301–307

Conference

https://doi.org/10.1007/978-3-032-17174-0_43

Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students

L. Motta, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, N. Busi

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 242–248

Conference

https://doi.org/10.1007/978-3-032-17174-0_34

Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform

H. Rojhan, F. Bellotti, **L. Lazzaroni**, M. Fresta, L. Forneris, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 218–223

Conference

https://doi.org/10.1007/978-3-032-17174-0_30

Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA

D. Faryar, R. Berta, M. Fresta, A. Saad, **L. Lazzaroni**, H. Ballout, O. Srour, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 25–31

Conference

https://doi.org/10.1007/978-3-032-17174-0_4

Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning

O.L. Audi, F. Bellotti, R. Berta, Z. Liu, **L. Lazzaroni**, A. Pilla, H. Rojhan, L. Forneris

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 212–217

Conference

https://doi.org/10.1007/978-3-032-17174-0_29

Implementing YoloV8 on Zynq UltraScale + MPSoC FPGA Boards

H. Ballout, R. Berta, O. Srour, **L. Lazzaroni**, M. Fresta, A. Saad, F. Bellotti

Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 289–294

National

https://doi.org/10.1007/978-3-032-17305-8_31

FPGA Implementation of a Robust Decision-Making System for Low-Speed Maneuvering in Automated Driving

A. Saad, H. Ballout, F. Bellotti, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta

Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 295–301

National

https://doi.org/10.1007/978-3-032-17305-8_32

Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision

A. Pighetti, F. Bellotti, S. Mohammady, **L. Lazzaroni**, H. Ballout, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 85–90

Conference

https://doi.org/10.1007/978-3-032-17174-0_12

Embedded Deployment of an LLM-Based Voice Assistant

L. Lazzaroni, F. Bellotti, A. Pighetti, A. Saad, O. Srour, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 273–278

Conference

https://doi.org/10.1007/978-3-032-17174-0_39

Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection

V. Soltanmuradov, F. Bellotti, R. Berta, D.M. Gómez, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 286–291

Conference

https://doi.org/10.1007/978-3-032-17174-0_41

A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors

S.G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, **L. Lazzaroni**, H. Ballout, H. Rojhan, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 183–190

Conference

https://doi.org/10.1007/978-3-032-17174-0_25

A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems

E. Castagno, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, S. Incerti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 249–254

Conference

https://doi.org/10.1007/978-3-032-17174-0_35

A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents

A. Zare, F. Bellotti, R. Berta, **L. Lazzaroni**, L. Forneris, A. Pighetti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 79–84

Conference

https://doi.org/10.1007/978-3-032-17174-0_11

2025

An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection

L. Lazzaroni, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, R. Berta

IEEE Sensors Journal vol. 25, no. 24, pp. 44859–44874

Journal

<https://doi.org/10.1109/jsen.2025.3627477>

TinyML Acceleration with MAX78000

A. Dabbous, **L. Lazzaroni**, F. Bellotti, S.M. Presentación, A. Pighetti, R. Berta

Proceedings of SIE 2024 Lecture Notes in Electrical Engineering pp. 468–474

National

https://doi.org/10.1007/978-3-031-71518-1_57

Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection

L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 156–163

Conference

https://doi.org/10.1007/978-3-031-84100-2_19

Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking

H. Ballout, R. Berta, I. Bochenko, **L. Lazzaroni**, A. Dhonthi, V. Hashemi, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 387–394

Conference

https://doi.org/10.1007/978-3-031-84100-2_46

Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking

F. Bellotti, R. Berta, V. Soltanmuradov, D.M. Gómez, A. Dhonthi, V. Hashemi, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 139–147

Conference

https://doi.org/10.1007/978-3-031-84100-2_17

RL-Based Generation of a Synthetic Automotive Driving Scenario Dataset

R. Berta, F. Bellotti, M. Cossu, L. Forneris, **L. Lazzaroni**, A. Pighetti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 395–402

Conference

https://doi.org/10.1007/978-3-031-84100-2_47

RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh

IEEE Sensors Letters vol. 9, no. 9, art. 6010204

Journal

<https://doi.org/10.1109/lensens.2025.3600982>

Quantization of MobileNetV2 for Resource-Constrained Microcontrollers

L. Lazzaroni, F. Bellotti, A. Dabbous, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 497–504

Conference

https://doi.org/10.1007/978-3-031-84100-2_59

Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board

H. Ballout, R. Berta, A. Dabbous, M. Fresta, **L. Lazzaroni**, A. Pighetti, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 242–249

Conference

https://doi.org/10.1007/978-3-031-84100-2_29

On-Board Deployability of a Deep Learning-Based System for Distraction and Inattention Detection

M. Fresta, F. Bellotti, **L. Lazzaroni**, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudeiro, F. Tango, R. Berta

IEEE Open Journal of the Industrial Electronics Society vol. 6, pp. 1351–1369

Journal

<https://doi.org/10.1109/ojies.2025.3601982>

Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices

A. Capello, R. Berta, M. Fresta, **L. Lazzaroni**, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 323–330

Conference

https://doi.org/10.1007/978-3-031-84100-2_38

Green Machine Learning for the IoT (GEMINI): A Position Paper

L. Lazzaroni, A. Dabbous, M. Nardello, M.A. Lebdeh, R. Berta, K.S. Yildirim, F. Bellotti, D. Brunelli

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 339–346

Conference

https://doi.org/10.1007/978-3-031-84100-2_40

Exploring Decision Transformer for Highway Automated Driving

L. Forneris, F. Bellotti, R. Berta, **L. Lazzaroni**, C. Oh

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 123–130

Conference

https://doi.org/10.1007/978-3-031-84100-2_15

Deep Learning-Based Real-Time Driver Cognitive Distraction Detection

M. Fresta, F. Bellotti, I. Bochenko, **L. Lazzaroni**, G. Merlhiot, F. Tango, R. Berta

IEEE Access vol. 13, pp. 26589–26607

Journal

<https://doi.org/10.1109/access.2025.3539392>

Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection

M. Cossu, R. Berta, M.A. Bagheri Orumi, **L. Lazzaroni**, J.L. Sauvaget, H. Touil, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 250–257

Conference

https://doi.org/10.1007/978-3-031-84100-2_30

Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices

A. Dabbous, **L. Lazzaroni**, F. Sakr, R. Berta, M. Fresta, F. Bellotti

2025 IEEE International Conference on Consumer Electronics (ICCE)

Digest of Technical Papers IEEE International Conference on Consumer Electronics

Conference

<https://doi.org/10.1109/icce63647.2025.10930047>

Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification

A. Dabbous, R. Berta, **L. Lazzaroni**, A. Pighetti, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 234–241

Conference

https://doi.org/10.1007/978-3-031-84100-2_28

2024

An embedded end-to-end voice assistant

L. Lazzaroni, F. Bellotti, R. Berta

Engineering Applications of Artificial Intelligence vol. 136, art. 108998

Journal

<https://doi.org/10.1016/j.engappai.2024.108998>

Adversarial Policy Generation in Automated Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh

ECCV 2024 Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving

Workshop

CBin-NN: An Inference Engine for Binarized Neural Networks

F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Bellotti

Electronics vol. 13, no. 9, art. 1624 Journal

<https://doi.org/10.3390/electronics13091624>

End-to-End Dataset Collection System for Sport Activities

M. Fresta, F. Bellotti, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Ansovini, R. Berta

Electronics vol. 13, no. 7, art. 1286 Journal

<https://doi.org/10.3390/electronics13071286>

YoloP-Based Pre-processing for Driving Scenario Detection

M. Cossu, R. Berta, L. Forneris, M. Fresta, **L. Lazzaroni**, J.L. Sauvaget, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 418–423 Conference

https://doi.org/10.1007/978-3-031-48121-5_60

Low-Cost, Edge-Cloud, End-to-End System Architecture for Human Activity Data Collection

M. Fresta, A. Dabbous, F. Bellotti, A. Capello, **L. Lazzaroni**, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 444–449 Conference

https://doi.org/10.1007/978-3-031-48121-5_64

Investigating Adversarial Policy Learning for Robust Agents in Automated Driving Highway Simulations

A. Pighetti, F. Bellotti, C. Oh, **L. Lazzaroni**, L. Forneris, M. Fresta, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 124–129 Conference

https://doi.org/10.1007/978-3-031-48121-5_18

Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity

R. Berta, **L. Lazzaroni**, A. Capello, M. Cossu, L. Forneris, A. Pighetti, F. Bellotti

Journal of Intelligent and Connected Vehicles vol. 7, no. 3, pp. 229–244 Journal

<https://doi.org/10.26599/jicv.2023.9210039>

Developing a TinyML Image Classifier in an Hour

R. Berta, A. Dabbous, **L. Lazzaroni**, D.P. Pau, F. Bellotti

IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 946–960 Journal

<https://doi.org/10.1109/ojies.2024.3451959>

Building a Pipeline for Efficient Production of Synthetic Datasets for Improving RL in Automated Driving

L. Lazzaroni, A. Pighetti, F. Bellotti, R. Berta

Proceedings of SIE 2023 Lecture Notes in Electrical Engineering pp. 351–356 National

https://doi.org/10.1007/978-3-031-48711-8_42

Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge

A. Dabbous, R. Berta, M. Fresta, H. Ballout, **L. Lazzaroni**, F. Bellotti

IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 781–794 Journal

<https://doi.org/10.1109/ojies.2024.3434341>

Automated Parking in CARLA: A Deep Reinforcement Learning-Based Approach

L. Lazzaroni, A. Pighetti, F. Bellotti, A. Capello, M. Cossu, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 352–357 Conference

https://doi.org/10.1007/978-3-031-48121-5_50

Application of a Serious Game for Emotion Elicitation Under Socio-Economic and Trust Based Decision-Making Scenarios for Autistic Adolescents

F. Ahmed, R. Berta, F. Bellotti, F. Floris, **L. Lazzaroni**, G. Barresi, J.R. Carrion

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 170–180

Conference

https://doi.org/10.1007/978-3-031-49065-1_17

2023

Implementing Deep Reinforcement Learning (DRL)-based Driving Styles for Non-Player Vehicles

L. Forneris, A. Pighetti, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, R. Berta

International Journal of Serious Games vol. 10, no. 4, pp. 153–170

Journal

<https://doi.org/10.17083/ijsg.v10i4.638>

Socio-Economic Decision Making and Emotion Elicitation with a Serious Game in the Wild

F. Ahmed, R. Berta, F. Bellotti, **L. Lazzaroni**, F. Floris, G. Barresi, J.R. Carrion

Applied Sciences vol. 13, no. 11, art. 6432

Journal

<https://doi.org/10.3390/app13116432>

Supporting a .csv-based Workflow in MongoDB for Data Analysts

M. Fresta, A. Capello, F. Bellotti, **L. Lazzaroni**, M. Cossu, R. Berta

2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE)

IEEE International Symposium on Industrial Electronics

Conference

<https://doi.org/10.1109/isie51358.2023.10228044>

Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 307–311

Conference

https://doi.org/10.1007/978-3-031-30333-3_41

Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

IEEE Access vol. 11, pp. 28522–28550

Journal

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Efficient Uploading of.Csv Datasets into a Non-Relational Database Management System

M. Fresta, F. Bellotti, A. Capello, M. Cossu, **L. Lazzaroni**, A. De Gloria, R. Berta

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Developing a Toolchain for Synthetic Driving Scenario Datasets

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

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Designing an Interpretability Analysis Framework for Deep Reinforcement Learning (DRL) Agents in Highway Automated Driving Simulation

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

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Deep Reinforcement Learning for Automated Car Parking

L. Lazzaroni, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

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Synthetic Driving Scenarios Generator

M. Cossu, R. Berta, A. Capello, A. De Gloria, L. Lazzaroni, F. Bellotti

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Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, L. Lazzaroni, M. Cossu, A. De Gloria, R. Berta

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Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, R. Berta, A. De Gloria, A. Capello, M. Cossu, L. Lazzaroni

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An Embedded End-to-End Voice Assistant

L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta

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Assessing Versatility of a Generic End-to-End Platform for IoT Ecosystem Applications

R. Berta, F. Bellotti, A. De Gloria, L. Lazzaroni

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Journal

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High-Level Decision-Making Non-player Vehicles

A. Pighetti, L. Forneris, L. Lazzaroni, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

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Developing a Synthetic Dataset for Driving Scenarios

J. Motta, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, L. Lazzaroni, S. Bonora

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Classifying Simulated Driving Scenarios from Automated Cars

M. Cossu, J.L.Q. Villon, F. Bellotti, A. Capello, A. De Gloria, L. Lazzaroni, R. Berta

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A Serious Game for Using Socio-Economic and Trust Based Decision-Making Scenarios for Elicitation of Emotional Responses

F. Ahmed, J.R. Carrion, F. Bellotti, **L. Lazzaroni**, G. Barresi, R. Berta

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Project and Development of an Italian Offline Embedded Voice Assistant

L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta

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Edgine, A Runtime System for IoT Edge Applications

R. Berta, A. Mazzara, F. Bellotti, A. De Gloria, **L. Lazzaroni**

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Adapting Autonomous Agents for Automotive Driving Games

G. Campodonico, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, T. Taccioli, F. Davio

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Employing an IoT Framework as a Generic Serious Games Analytics Engine

L. Lazzaroni, A. Mazzara, F. Bellotti, A. De Gloria, R. Berta

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